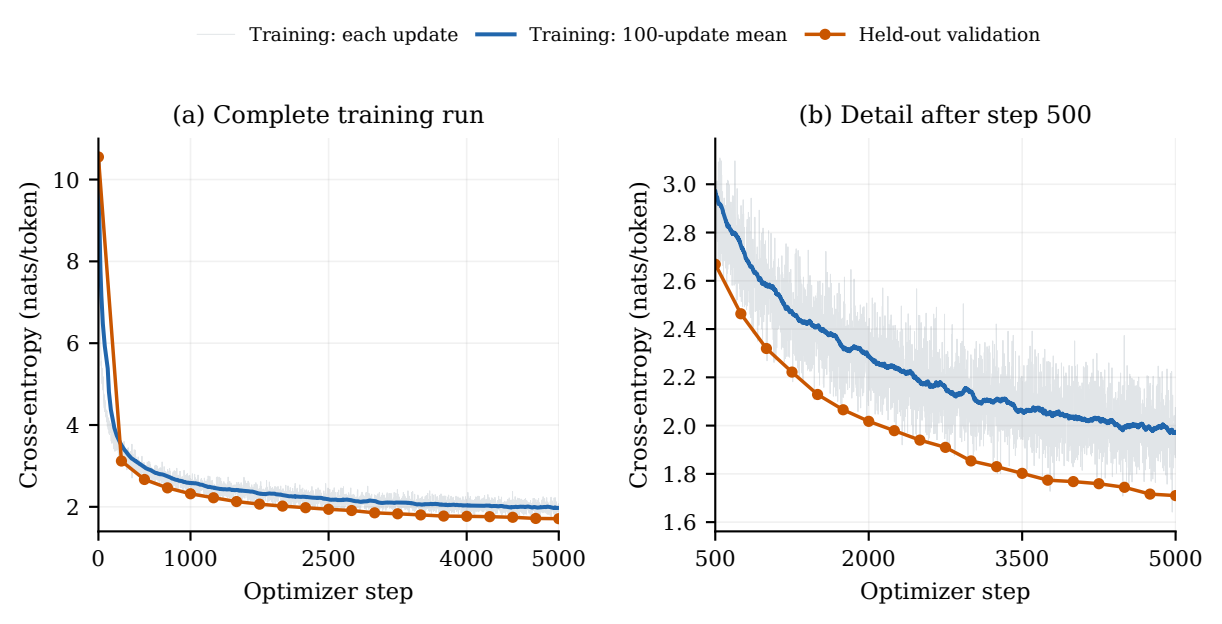

Training: each update
Training: 100-update mean
Held-out validation
(a) Complete training run
(b) Detail after step 500
Cross-entropy (nats/token)
Optimizer step
10
8
6
4
2
0
1000
2500
4000
5000
3.0
2.8
2.6
2.4
2.2
2.0
1.8
1.6
500
2000
3500